

# Eisenstein Sources for the Twelve Modular Sporadic Apéry Companions

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## Abstract

For the twelve sporadic Apéry-like recurrences admitting modular parametrizations, we prove that the canonical source of the normalized second solution is a pure Eisenstein series. The key simplification is the identity

$$\Phi(\tau) = F(\tau) \theta_q t(\tau),$$

where  $t$  is the canonical nome coordinate and  $F$  is the analytic solution pulled back to the modular parameter. Eleven cases are proved uniformly. Their candidate  $t$  and  $F$  are eta quotients; Ligozat's criterion proves modularity, and two small holomorphic modular identities — a normalized Wronskian identity and a Rankin–Cohen form of the differential equation — prove that these eta quotients are the *canonical* recurrence parametrizations. Sturm's theorem then identifies  $F\theta_q t$  with the stated Eisenstein series. The remaining Apéry  $\zeta(2)$  family is treated on  $\Gamma_1(5)$  using Zagier's Rogers–Ramanujan parametrization and a weight-six square identity. Thus the previously finite coefficient identifications become identities for all Fourier coefficients, and no cuspidal component occurs in any of the twelve sources.

## 1 The recurrences and the source

Put  $\theta_t = t \, d/dt$  and  $\theta_q = q \, d/dq$ . We use the two operator families

$$L_2 = \theta_t^2 - t(a\theta_t^2 + a\theta_t + b) + ct^2(\theta_t + 1)^2, \quad (1)$$

$$L_3 = \theta_t^3 - t(2\theta_t + 1)(a\theta_t^2 + a\theta_t + b) + ct^2(\theta_t + 1)^3. \quad (2)$$

Their leading polynomials are

$$P_2(t) = 1 - at + ct^2, \quad P_3(t) = 1 - 2at + ct^2.$$

Let  $y_0(t) = 1 + O(t)$  be the analytic solution, let  $q$  be the canonical Frobenius nome, and write

$$t = t(q), \quad F(q) = y_0(t(q)), \quad K(q) := \theta_q \log t = \frac{\theta_q t}{t}.$$

Thus  $t = q + O(q^2)$  and  $K = 1 + O(q)$ .

The normalized companion is the unique log-free series  $y_B(t)$  with  $y_B(0) = 0$  and  $L_r y_B = t$ . Exact rectification by the canonical nome gives

$$L_r = C(t) F \theta_q^r F^{-1}.$$

Since  $\theta_q = K\theta_t$ , comparison of the coefficient of  $\theta_t^r$  gives

$$C(t) = \frac{P_r(t)}{K^r}. \quad (3)$$

Consequently

$$y_B = F\theta_q^{-r}\Phi, \quad \Phi = \frac{tK^r}{P_r F}. \quad (4)$$

For  $r = 2$  rectification is automatic from the Frobenius basis. For  $r = 3$  we use the symmetric-square identity of Almkvist–Zudilin [1, Prop. 9]:

$$\tilde{L} = \theta_t^2 - t \left( 2a\theta_t^2 + a\theta_t + \frac{b}{2} \right) + ct^2 \left( \theta_t + \frac{1}{2} \right)^2, \quad (5)$$

whose symmetric square is  $L_3$ . If  $u_0, u_1$  are its Frobenius solutions, then  $F = u_0^2$  and  $u_1/u_0 = \log q$ .

**Proposition 1.1** (collapse of the source). *For both operator families, in the canonical coordinate,*

$$\boxed{\Phi = F \theta_q t}. \quad (6)$$

*Proof.* For  $L_2$ , the Wronskian of  $F$  and  $F \log q$  is

$$F^2 \frac{d \log q}{dt} = \frac{1}{t P_2(t)},$$

with normalization fixed at  $t = 0$ . Hence  $K = P_2 F^2$ . Substitution into (4) gives

$$\Phi = \frac{t(P_2 F^2)^2}{P_2 F} = t P_2 F^3 = F \theta_q t.$$

For  $L_3$ , the Wronskian of the two solutions of (5) is  $1/(t\sqrt{P_3})$ . Since  $F = u_0^2$ , this gives  $K = F\sqrt{P_3}$ , equivalently  $K^2 = P_3 F^2$ . Therefore

$$\Phi = \frac{tK^3}{P_3 F} = tKF = F \theta_q t.$$

□

The point of (6) is that the source is no longer an opaque quotient:  $\theta_q t$  is a meromorphic modular form of weight 2, so a modular  $F$  of weight  $r - 1$  makes  $\Phi$  a form of weight  $r + 1$ .

## 2 A uniform finite certificate for eleven eta families

Write  $\eta_d = \eta(d\tau)$ , with  $q = e^{2\pi i\tau}$ . For an eta quotient  $G = \prod_{d|N} \eta_d^{e_d}$  we use the standard Ligozat cusp-order formula

$$\text{ord}_{1/c} G = \frac{N}{24c \gcd(c, N/c)} \sum_{d|N} \frac{\gcd(c, d)^2}{d} e_d, \quad c \mid N. \quad (7)$$

The usual two congruences  $\sum d e_d \equiv \sum (N/d) e_d \equiv 0 \pmod{24}$  and (7) certify the forms below. Every displayed  $t$  has weight zero, trivial character, and Fourier expansion  $q + O(q^2)$ ; every  $F$  is holomorphic, has constant term 1, and has the character stated below.

It remains important to prove that Table 1 gives the *canonical* parametrizations of the recurrences, rather than modular forms whose first few coefficients happen to agree. This is done by two small modular identities.

Let  $D = \theta_q$  and  $K = D \log t$ . For an eta quotient,

$$K = \frac{1}{24} \sum_{d|N} e_d d E_2(d\tau), \quad (8)$$

| family        | $N$ | $\chi_F$    | $t$                                                                     | $F$                                                                  |
|---------------|-----|-------------|-------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>A</b>      | 6   | $\chi_{-3}$ | $\eta_1^3 \eta_6^9 / (\eta_2^3 \eta_3^9)$                               | $\eta_2 \eta_3^6 / (\eta_1^2 \eta_6^3)$                              |
| <b>B</b>      | 36  | $\chi_{-3}$ | $\eta_1^3 \eta_4^3 \eta_{18}^9 / (\eta_2^9 \eta_3^3 \eta_{36}^3)$       | $\eta_2^9 \eta_3 \eta_{12} / (\eta_1^3 \eta_4^3 \eta_6^3)$           |
| <b>C</b>      | 6   | $\chi_{-3}$ | $\eta_1^4 \eta_6^8 / (\eta_2^8 \eta_3^4)$                               | $\eta_2^6 \eta_3 / (\eta_1^3 \eta_6^2)$                              |
| <b>E</b>      | 8   | $\chi_{-4}$ | $\eta_1^4 \eta_2^2 \eta_8^4 / \eta_2^{10}$                              | $\eta_2^{10} / (\eta_1^4 \eta_4^4)$                                  |
| <b>F</b>      | 12  | $\chi_{-3}$ | $\eta_1^5 \eta_3 \eta_4^5 \eta_6^2 \eta_{12} / \eta_2^{14}$             | $\eta_2^{15} \eta_3^2 \eta_{12}^2 / (\eta_1^6 \eta_4^6 \eta_6^5)$    |
| $\alpha$      | 12  | 1           | $\eta_1^6 \eta_3^6 \eta_4^6 \eta_{12}^6 / (\eta_2^{12} \eta_6^{12})$    | $\eta_2^{10} \eta_6^{10} / (\eta_1^4 \eta_3^4 \eta_4^4 \eta_{12}^4)$ |
| $\gamma$      | 6   | 1           | $\eta_1^{12} \eta_6^{12} / (\eta_2^{12} \eta_3^{12})$                   | $\eta_2^7 \eta_3^7 / (\eta_1^5 \eta_6^5)$                            |
| $\delta$      | 12  | 1           | $\eta_1^4 \eta_4^4 \eta_6^{16} / (\eta_2^{16} \eta_3^4 \eta_{12}^4)$    | $\eta_2^{12} \eta_3 \eta_{12} / (\eta_1^3 \eta_4^3 \eta_6^4)$        |
| $\varepsilon$ | 8   | 1           | $\eta_1^8 \eta_8^8 / (\eta_2^8 \eta_4^8)$                               | $\eta_2^6 \eta_4^6 / (\eta_1^4 \eta_8^4)$                            |
| $\zeta$       | 9   | 1           | $\eta_1^6 \eta_9^6 / \eta_3^{12}$                                       | $\eta_3^{10} / (\eta_1^3 \eta_9^3)$                                  |
| $\eta$        | 20  | $\chi_5$    | $\eta_1^6 \eta_4^6 \eta_{10}^{18} / (\eta_2^{18} \eta_5^6 \eta_{20}^6)$ | $\eta_2^{15} \eta_5 \eta_{20} / (\eta_1^5 \eta_4^5 \eta_{10}^3)$     |

Table 1: The eleven ordinary eta parametrizations. The first five rows have  $\text{wt } F = 1$ ; the last six have  $\text{wt } F = 2$ .

and, because  $t$  is a weight-zero modular function,  $K$  is a genuine holomorphic weight-two modular form. Define

$$H_2 = FD^2F - 2(DF)^2 + t(-b + ct)P_2(t)F^6, \quad (9)$$

$$H_3 = 2FD^2F - 3(DF)^2 + t(-2b + ct)F^4. \quad (10)$$

The derivative combinations are Rankin–Cohen brackets, hence modular and holomorphic. Formula (7) shows that at every pole of  $t$  we have

$$\text{ord } F = -\text{ord } t. \quad (11)$$

Thus all terms in the identities below are holomorphic at every cusp.

**Lemma 2.1** (Wronskian–ODE certificate). *For a row of type R2, if*

$$K = P_2F^2, \quad H_2 = 0, \quad (12)$$

*then  $F(t(q))$  is the normalized analytic solution of  $L_2$ , and  $q$  is its canonical nome. For a row of type R3, if*

$$K^2 = P_3F^2, \quad H_3 = 0, \quad (13)$$

*then  $F = u_0^2$ , where  $u_0$  is the normalized analytic solution of (5), and again  $q$  is the canonical nome of  $L_3$ .*

*Proof.* For R2, write  $A = tP_2'$  and  $R = t(-b + ct)$ . Substituting  $\theta_t = K^{-1}D$  into  $L_2F$  and using  $K = P_2F^2$  gives the exact identity

$$K^3L_2F = P_2^2F H_2.$$

Thus  $L_2F = 0$ . Moreover  $K = P_2F^2$  is equivalent to

$$\theta_t \log q = \frac{1}{P_2F^2},$$

which is precisely the normalized reduction-of-order formula for the ratio of the logarithmic Frobenius solution to  $F$ . Since  $t = q + O(q^2)$ , the additive constant is fixed and  $q$  is the canonical nome.

For  $R3$ , put  $u = F^{1/2} = 1 + O(q)$  and use  $K^2 = P_3 F^2$ , taking the branch  $K = F\sqrt{P_3}$  at  $q = 0$ . A direct substitution in (5) yields

$$\frac{\tilde{L}u}{u} = \frac{H_3}{4F^4}.$$

Hence  $u$  solves the root equation. The identity  $\theta_t \log q = (F\sqrt{P_3})^{-1}$  is its normalized reduction-of-order formula, so  $q$  is canonical. The symmetric-square relation gives  $F = u^2$  as the analytic solution of  $L_3$ .  $\square$

The identities in Lemma 2.1, and then the source identity, are now finite Sturm checks. The parameters are

| family        | type | $(a, b, c)$  | level | wt $F$ |
|---------------|------|--------------|-------|--------|
| <b>A</b>      | R2   | (7, 2, -8)   | 6     | 1      |
| <b>B</b>      | R2   | (9, 3, 27)   | 36    | 1      |
| <b>C</b>      | R2   | (10, 3, 9)   | 6     | 1      |
| <b>E</b>      | R2   | (12, 4, 32)  | 8     | 1      |
| <b>F</b>      | R2   | (17, 6, 72)  | 12    | 1      |
| $\alpha$      | R3   | (10, 4, 64)  | 12    | 2      |
| $\gamma$      | R3   | (17, 5, 1)   | 6     | 2      |
| $\delta$      | R3   | (7, 3, 81)   | 12    | 2      |
| $\varepsilon$ | R3   | (12, 4, 16)  | 8     | 2      |
| $\zeta$       | R3   | (9, 3, -27)  | 9     | 2      |
| $\eta$        | R3   | (11, 5, 125) | 20    | 2      |

| family        | Wronskian bound | ODE bound | source bound | poles of $t$                 |
|---------------|-----------------|-----------|--------------|------------------------------|
| <b>A</b>      | 2               | 6         | 3            | $1/3 : -1$                   |
| <b>B</b>      | 12              | 36        | 18           | $1 : -1, 1/2 : -4, 1/4 : -1$ |
| <b>C</b>      | 2               | 6         | 3            | $1/2 : -1$                   |
| <b>E</b>      | 2               | 6         | 3            | $1/2 : -1$                   |
| <b>F</b>      | 4               | 12        | 6            | $1/2 : -2$                   |
| $\alpha$      | 8               | 16        | 8            | $1/2 : -2, 1/6 : -2$         |
| $\gamma$      | 4               | 8         | 4            | $1/2 : -1, 1/3 : -1$         |
| $\delta$      | 8               | 16        | 8            | $1 : -1, 1/2 : -2, 1/4 : -1$ |
| $\varepsilon$ | 4               | 8         | 4            | $1/2 : -1, 1/4 : -1$         |
| $\zeta$       | 4               | 8         | 4            | $1/3 : -1$                   |
| $\eta$        | 12              | 24        | 12           | $1 : -1, 1/2 : -4, 1/4 : -1$ |

Table 2: Sturm bounds for the three exact identities. A pole entry  $1/c : -m$  means  $\text{ord}_{1/c} t = -m$ ; in every entry  $\text{ord}_{1/c} F = m$ , establishing (11).

**Proposition 2.2.** *For every row of Table 1, the displayed  $t, F$  are the canonical parametrization of the corresponding recurrence.*

*Proof.* Ligozat's criterion gives the stated weights, characters and cusp orders. The two sides of the Wronskian identity in (12) or (13) are holomorphic modular forms of weight 2 or 4, respectively. The forms  $H_2, H_3$  are holomorphic of weight 6 and 8. Exact rational Fourier arithmetic verifies the identities through the bounds in the first two numerical columns of Table 2. Sturm's theorem therefore proves them identically. Lemma 2.1 finishes the proof.  $\square$

### 3 The Eisenstein identities

For a Dirichlet character  $\chi$  put

$$\sigma_\chi^{(k)}(m) = \sum_{d|m} \chi(m/d) d^k, \quad \tilde{\sigma}_\chi^{(k)}(m) = \sum_{d|m} \chi(d) d^k,$$

and let  $\sigma_3(m) = \sum_{d|m} d^3$ . A value at a non-integer argument, such as  $\sigma_3(m/2)$ , is understood as zero unless the denominator divides  $m$ .

| family        | coefficient $c(m)$ of $E = \sum_{m \geq 1} c(m) q^m$                                                      |
|---------------|-----------------------------------------------------------------------------------------------------------|
| <b>A</b>      | $\tilde{\sigma}_{\chi_{-3}}^{(2)}(m) - \tilde{\sigma}_{\chi_{-3}}^{(2)}(m/2)$                             |
| <b>B</b>      | $\sigma_{\chi_{-3}}^{(2)}(m) - 6\sigma_{\chi_{-3}}^{(2)}(m/2) - 8\sigma_{\chi_{-3}}^{(2)}(m/4)$           |
| <b>C</b>      | $\sigma_{\chi_{-3}}^{(2)}(m) - 8\sigma_{\chi_{-3}}^{(2)}(m/2)$                                            |
| <b>E</b>      | $\sigma_{\chi_{-4}}^{(2)}(m) - 8\sigma_{\chi_{-4}}^{(2)}(m/2)$                                            |
| <b>F</b>      | $\sigma_{\chi_{-3}}^{(2)}(m) - 7\sigma_{\chi_{-3}}^{(2)}(m/2) - 8\sigma_{\chi_{-3}}^{(2)}(m/4)$           |
| $\alpha$      | $\sigma_3(m) - 17\sigma_3(m/2) - 9\sigma_3(m/3) + 16\sigma_3(m/4) + 153\sigma_3(m/6) - 144\sigma_3(m/12)$ |
| $\gamma$      | $\sigma_3(m) - 28\sigma_3(m/2) + 63\sigma_3(m/3) - 36\sigma_3(m/6)$                                       |
| $\delta$      | $\sigma_3(m) - 14\sigma_3(m/2) - \sigma_3(m/3) + 16\sigma_3(m/4) + 14\sigma_3(m/6) - 16\sigma_3(m/12)$    |
| $\varepsilon$ | $\sigma_3(m) - 21\sigma_3(m/2) + 84\sigma_3(m/4) - 64\sigma_3(m/8)$                                       |
| $\zeta$       | $\sum_{d m} \chi_{-3}(d) \chi_{-3}(m/d) d^3$                                                              |
| $\eta$        | $\sigma_{\chi_5}^{(3)}(m) - 14\sigma_{\chi_5}^{(3)}(m/2) - 16\sigma_{\chi_5}^{(3)}(m/4)$                  |

Table 3: The eleven eta-family Eisenstein sources.

Each row is a standard Eisenstein series or a linear combination of its oldform embeddings. In the ordinary  $E_4$  rows the coefficients of the oldforms sum to zero, so the constant term cancels. In the character rows the constant term is either zero by primitivity or cancels between the displayed oldforms. Thus Table 3 defines a holomorphic form in the same  $M_{r+1}(\Gamma_0(N), \chi_F)$  as  $F\theta_q t$ .

**Proposition 3.1.** *For all eleven rows of Table 1,*

$$\boxed{F\theta_q t = E},$$

where  $E$  is the Eisenstein series of Table 3.

*Proof.* At a pole of  $t$ , equation (11) shows that  $tF$  has order zero, while  $K = D \log t$  is holomorphic. Hence  $F\theta_q t = tFK$  is holomorphic at every cusp; it is also holomorphic in the upper half-plane because eta has no zeros there. Its weight is  $r+1$  and its character is  $\chi_F$ . Exact Fourier arithmetic gives  $F\theta_q t - E = 0$  through the source Sturm bound in Table 2; Sturm's theorem proves the identity.  $\square$

### 4 The D family on $\Gamma_1(5)$

The remaining order-two sporadic family has  $(a, b, c) = (11, 3, -1)$  and  $P(t) = 1 - 11t - t^2$ . It is the classical Apéry  $\zeta(2)$  family. Zagier's exact modular parametrization [5] gives

$$t = q \prod_{n \geq 1} (1 - q^n)^{5(\frac{n}{5})}, \quad P(t) = t \frac{\eta_1^6}{\eta_5^6}, \quad F^2 = t^{-1} \frac{\eta_5^5}{\eta_1}. \quad (14)$$

The corresponding Beauville family is the one with group  $\Gamma_1(5)$ . By Proposition 1.1,  $\Phi = F\theta_q t = tPF^3$ , and (14) therefore gives

$$\boxed{\Phi^2 = t\eta_1^9\eta_5^3.} \quad (15)$$

Let  $\chi$  be the primitive quartic character modulo 5, normalized by  $\chi(2) = i$ . Put  $\psi_1 = \Re\chi$ ,  $\psi_2 = \Im\chi$  and

$$E_D = \sum_{m \geq 1} \left( \sum_{d|m} (\psi_1(d) - 2\psi_2(d))d^2 \right) q^m. \quad (16)$$

For clarity, let  $\mathcal{E}_\chi$  denote the standard weight-three Eisenstein series whose positive Fourier coefficients are  $\sum_{d|m} \chi(d)d^2$ . Then

$$E_D = \left( \frac{1}{2} + i \right) \mathcal{E}_\chi + \left( \frac{1}{2} - i \right) \mathcal{E}_{\bar{\chi}}.$$

Both summands are holomorphic modular forms on  $\Gamma_1(5)$ . With the standard normalization  $B_{3,\chi} = (12 + 6i)/5$ , their constant terms cancel exactly, so (16) is a holomorphic weight-three Eisenstein form with zero constant term.

**Lemma 4.1.** *The right-hand side of (15) is a holomorphic weight-six form on  $\Gamma_1(5)$ .*

*Proof.* The eta product  $G = \eta_1^9\eta_5^3$  is a weight-six form on  $\Gamma_0(5)$  with quadratic character  $\chi_5$ , which becomes trivial on  $\Gamma_1(5)$ . Its cusp orders on  $X_0(5)$  are 2 and 1. The modular function  $h = \eta_1^6/\eta_5^6$  has corresponding orders 1 and  $-1$ . If a cusp of  $X_1(5)$  above the pole cusp of  $h$  has ramification index  $e$ , then (14) and  $P(t) \sim -t^2$  force  $\text{ord } t = -e$ , whereas  $\text{ord } G = e$ ; the pole cancels. Above the other cusp, where  $h$  has a zero, the same order comparison shows that  $t$  cannot have a pole. There are no poles in the upper half-plane, since  $h$  is nonzero there and  $P(t) = th$  would otherwise have incompatible pole orders. Thus  $tG$  is holomorphic.  $\square$

**Proposition 4.2.** *For the  $\mathbf{D}$  family,  $\Phi = E_D$ .*

*Proof.* Both  $\Phi^2 = t\eta_1^9\eta_5^3$  and  $E_D^2$  lie in  $M_6(\Gamma_1(5))$ . A safe Sturm bound is

$$\frac{6}{12}[\text{SL}_2(\mathbb{Z}) : \Gamma_1(5)] = \frac{6}{12} \cdot 24 = 12.$$

The exact expansion

$$E_D = q - 7q^2 + 19q^3 - 23q^4 + q^5 + 47q^6 - 97q^7 + 105q^8 - 62q^9 - 7q^{10} + 122q^{11} - 257q^{12} + O(q^{13})$$

gives

$$\begin{aligned} E_D^2 &= q^2 - 14q^3 + 87q^4 - 312q^5 + 685q^6 - 794q^7 - 285q^8 \\ &\quad + 3308q^9 - 7441q^{10} + 9400q^{11} - 4829q^{12} + O(q^{13}). \end{aligned}$$

Direct expansion of the product on the right of (15) gives exactly the same coefficients. Sturm's theorem yields  $\Phi^2 = E_D^2$ . Since both  $\Phi$  and  $E_D$  begin with  $q$ , the sign is positive:  $\Phi = E_D$ .  $\square$

## 5 Main theorem and scope

**Theorem 5.1** (Eisenstein-source theorem). *For each of the twelve modularly identifiable sporadic Apéry-like families*

$$\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}, \mathbf{E}, \mathbf{F}, \alpha, \gamma, \delta, \varepsilon, \zeta, \eta,$$

*the normalized companion source is*

$$\Phi = F\theta_q t,$$

*and is exactly the pure Eisenstein series specified in Table 3, with the  $\mathbf{D}$  row given by (16). In particular, the cuspidal projection of every one of these twelve sources is zero.*

*Proof.* Proposition 1.1 identifies the canonical source. Propositions 2.2 and 3.1 prove the eleven eta cases. Proposition 4.2 proves  $\mathbf{D}$ .  $\square$

*Remark 5.2* (Literature and novelty scope). The second-order side overlaps substantially with Zagier’s period analysis: for the six sporadic  $R2$  recurrences, weight-three Eisenstein series and their Eichler integrals already occur explicitly in his modular treatment [5]. We therefore do not claim the discovery of Eisenstein behaviour for those individual families. The contribution here is the uniform identification of the *canonical companion source*  $\Phi = F\theta_q t$ , the exact twelve-family source table in one normalization, and a single finite certification that also proves canonicity of the eta parametrizations. Case-by-case priority beyond the cited classical examples should be assessed separately.

The theorem does not assert anything about the three additional Cooper rows  $s_7, s_{10}, s_{18}$  in their canonical coordinates. Their canonical nomes are integral, but the eta parametrizations used above are not available in that normalization. Nor does the theorem by itself evaluate every fold connection value: it identifies the modular source exactly. The subsequent passage from an Eisenstein source to a zeta or Dirichlet  $L$ -value is an Eichler-integral/connection problem and should be kept logically separate.

## 6 Exact certificate

All finite checks in Tables 2 and 3 were made with exact rational arithmetic. The supplementary script `eisenstein_source_certificate_v2.py` constructs the eta products from finite Euler products, computes  $K$  from (8), checks the Ligozat congruences and cusp orders, and verifies the Wronskian, ODE and source identities through their stated Sturm bounds. It also performs the  $\Gamma_1(5)$  square check through  $q^{12}$ . As an implementation guard, the source identities are additionally checked through  $q^{60}$ ; those extra coefficients are not used in the proofs.

## References

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